

Phase 5: Apex Programming

Project: SmartCare CRM (Healthcare Management System)

This phase introduces server-side automation using **Apex**, Salesforce's native programming language. The goal is to create a simple Apex Trigger that enforces data integrity on new appointment records, ensuring every appointment has a valid reason before it can be saved. This demonstrates how developer tools enhance the application's functionality.

1. Apex Trigger: Appointment Reason Validation

Purpose:

To ensure that every new appointment record includes a reason for the visit. This prevents users from creating incomplete records and improves data quality for doctors and administrators.

How it Works:

This is a **before insert trigger**, which runs just before a new Appointment__c record is saved to the database. The trigger checks each new appointment, and if the Reason__c field is empty, it throws an error message to the user, stopping the save operation.

Implementation and Testing

A. The Apex Code

The following trigger was created using the Developer Console and saved in Salesforce:



```
Developer Console - Google Chrome
orgfarm-b14977b716-dev-ed.develop.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage
File Edit Debug Test Workspace Help < >
AppointmentTrigger.apxt
Code Coverage: None API Version: 64 Go To

1 trigger AppointmentTrigger on Appointment__c (before insert) {
2   for (Appointment__c appt : Trigger.new) {
3     if (String.isEmpty(appt.Reason__c)) {
4       appt.addError('Reason is required for the appointment.');
```

B. Testing the Trigger

1. While Creating the Appointment

A new appointment record was created without entering a value in the **Reason** field.

2. After Clicking Save

The trigger fired successfully. Salesforce displayed the error message:

“Reason is required for the appointment.”

The record was not saved until a valid reason was entered.

The screenshot displays the 'New Appointment: New Consultation' form in Salesforce. The form is divided into two main sections: 'Patient Appointments' and 'Clinical Information'. In the 'Patient Appointments' section, the following fields are visible: '* Patient' (PAT-0004), '* Assigned Doctor' (Arun), '* Appointment DateTime' (Date: 9/25/2025, Time: 12:00 PM), '* Status' (Requested), '* Visit Type' (In-Person), and 'Consultation Fee'. A 'Reminder Sent' checkbox is also present. In the 'Clinical Information' section, the 'Reason' field is visible but empty. A red error message box is overlaid on the form, stating 'We hit a snag.' and 'Review the errors on this page.' with a bullet point: 'Reason is required for the appointment.' The form has 'Cancel', 'Save & New', and 'Save' buttons at the bottom. The background shows a list of recent appointments.

3. After giving a reason

The trigger is not fired and new appointment record is created with reason.

SmartCare Hospital

PatientsAppointmentsDoctorsPrescriptionsReportsDashboards

Appointment
APT-0009

New ContactEditNew Opportunity

RelatedDetails

Patient Appointments

Patient	Assigned Doctor
PAT-0004	Arun
Appointment DateTime	Status
9/25/2025, 12:00 PM	Requested
Visit Type	Consultation Fee
In-Person	
Reminder Sent	
☐	

Clinical Information

Reason	Owner
For Slight Pain in the left side of chest	Hospital Admin
Appointment Number	Last Modified By
APT-0009	Hospital Admin , 9/25/2025, 11:27 AM
Created By	
Hospital Admin , 9/25/2025, 11:27 AM	

Activity

Filters: All time • All activities • All types

RefreshExpand AllView All

Upcoming & Overdue

No activities to show.

Get started by sending an email, scheduling a task, and more.

No past activity. Past meetings and tasks marked as done show up here.

Conclusion

The Apex Trigger was successfully created, tested, and deployed. This enforces business rules at the database level and ensures data quality. With this, the SmartCare CRM project meets the Phase 5 requirement for introducing developer-level customizations using Apex.